

1. Implementation Summary

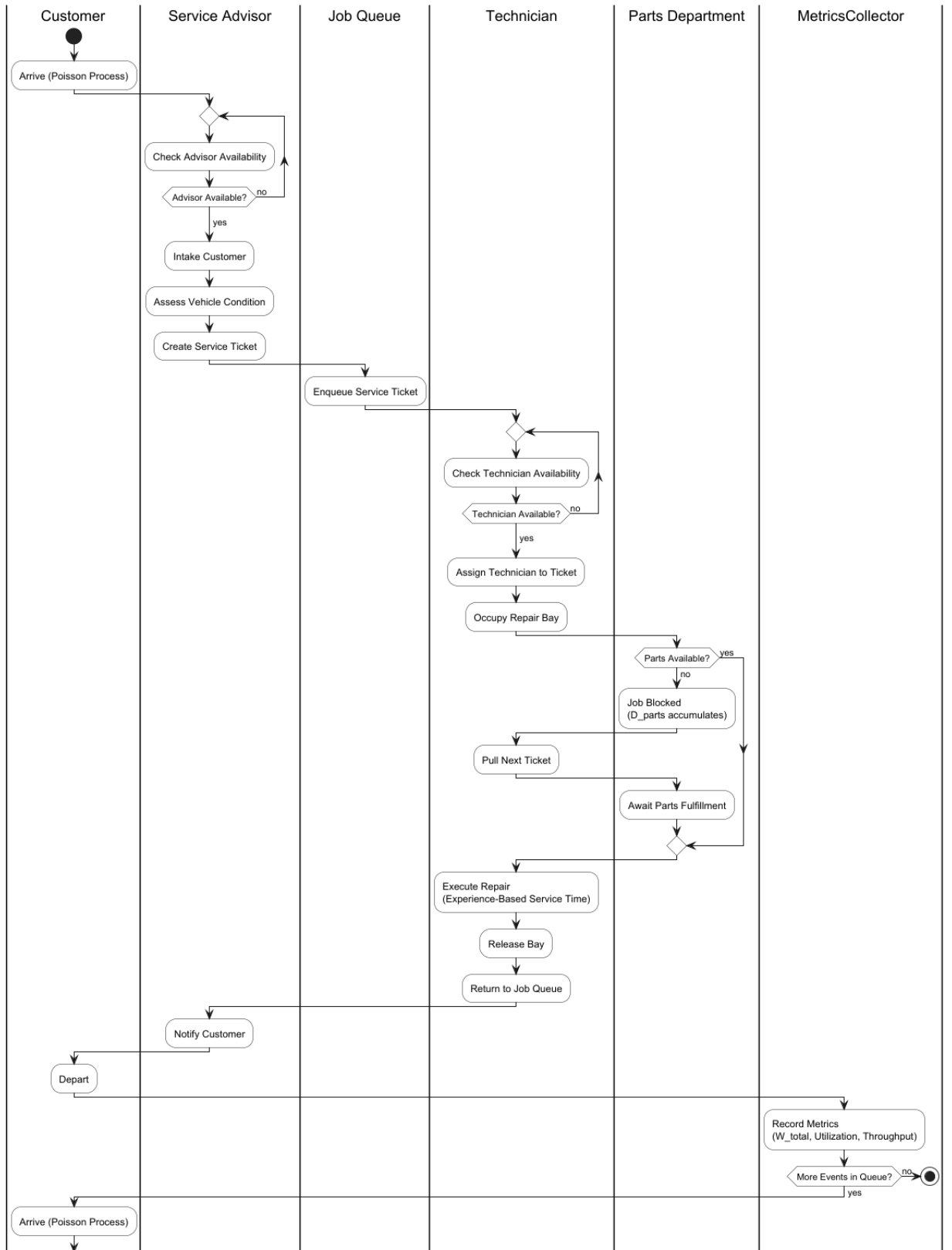
The Service Department Operational Optimization Simulator was designed to be a discrete-event simulation of a dealership's service department. The intended use-case is to assist management. It models how a customer's vehicle will move the service department identifying any potential bottlenecks, technician queue times, parts supply and shop utilization for management. The software gives data on customer wait times, utilization on various entities, throughput, and delays based on various user modifiable inputs.

Feature	Status	Implementation
DES engine	Complete	SimulationEngine- priority queue scheduler
Customer arrival process	Complete	CoxProcess and PiecewiseConstantArrivalRate
Service advisor intake	Complete	Constant 0.1-hour intake per customer
Entity Lifecycle	Complete	Customer > ticket > queue > technician > completion
Experience based service times	Complete	Gamma distribution scaled based on experience
Job Queuing	Complete	FIFO within JobQueue
Parts Inventorying	Complete	PartsDepartment
Metrics Collection	Complete	MetricsCollector
Analytics	Complete	SimulationValidationMetrics-Erlang C M/M/c
Parameterization	Complete	Simulation.properties, CLI flags, & SimulationConfig
Data Exports	Complete	RunResultWriter
UI	Complete	LandingView, ConfigView & ResultsView

A basic flow for the simulation is that:

- a. Customer arrives (Simulation uses a Cox Process)

- b. Once an advisor opens up, the advisor creates a ticket for the customer
(simulated 0.1 hours per ticket creation)
- c. A part check happens
 - i. Checks if at least 40% parts availability is there, if not then ticket becomes blocked until parts become available.
- d. Tickets enter the waiting queue for a technician to pick it up
- e. Technician gets assigned a bay and ticket
- f. Service time gets sampled (by the Gamma process plus experience)
- g. Job finishes and the metrics get recorded



Parameters

Parameter	Default	Config key
Arrival rate	4.0	--arrival-rate
Technicians	3	--technicians
Service rate	2.0	--service-rate
Advisors	4	--advisors
Seed	42	--seed
Gamma shape k	4.0	--gamma-shape
Experience α	1.0	--experience-alpha
Parts reorder point	5	--reorder-point
Parts lead time	2.0 h	--parts-lead-time
Validation tolerance	0.15	--validation-tolerance
Horizon	10 h	--horizon

Software Changes

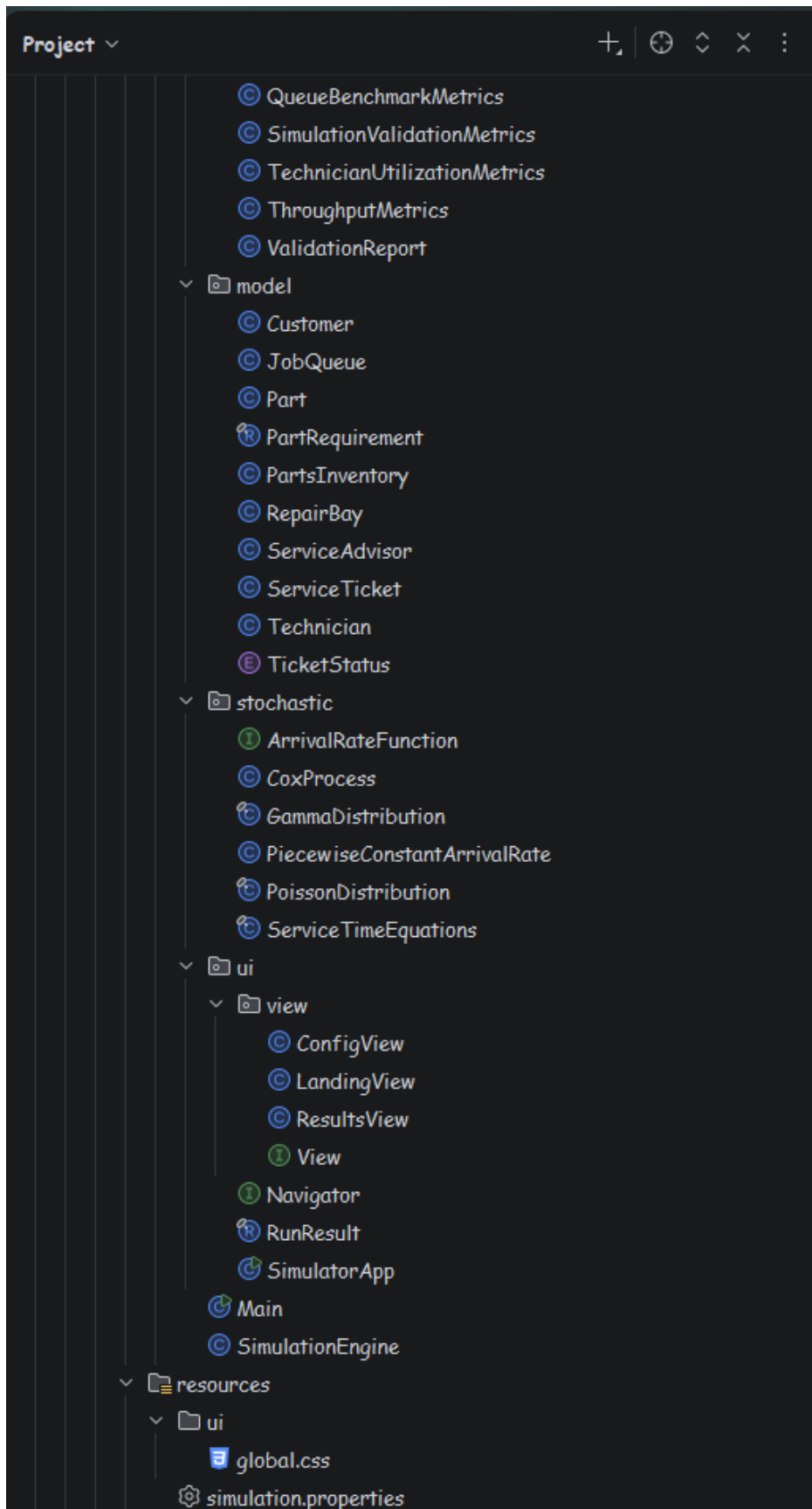
Original plan	Current Implementation	Reason for the change
SimKit library	Built-In scheduler	Easier to maintain
Java 22	Java 17	More mature library selection
Advisor Experience does affect intake times	Constant intake time	Matrches assumptions
Parameters were set as TBD	Concrete properties were set	Needs runnable simulation
Financial Modelling	Not implemented	Out of scope for project

Design updates

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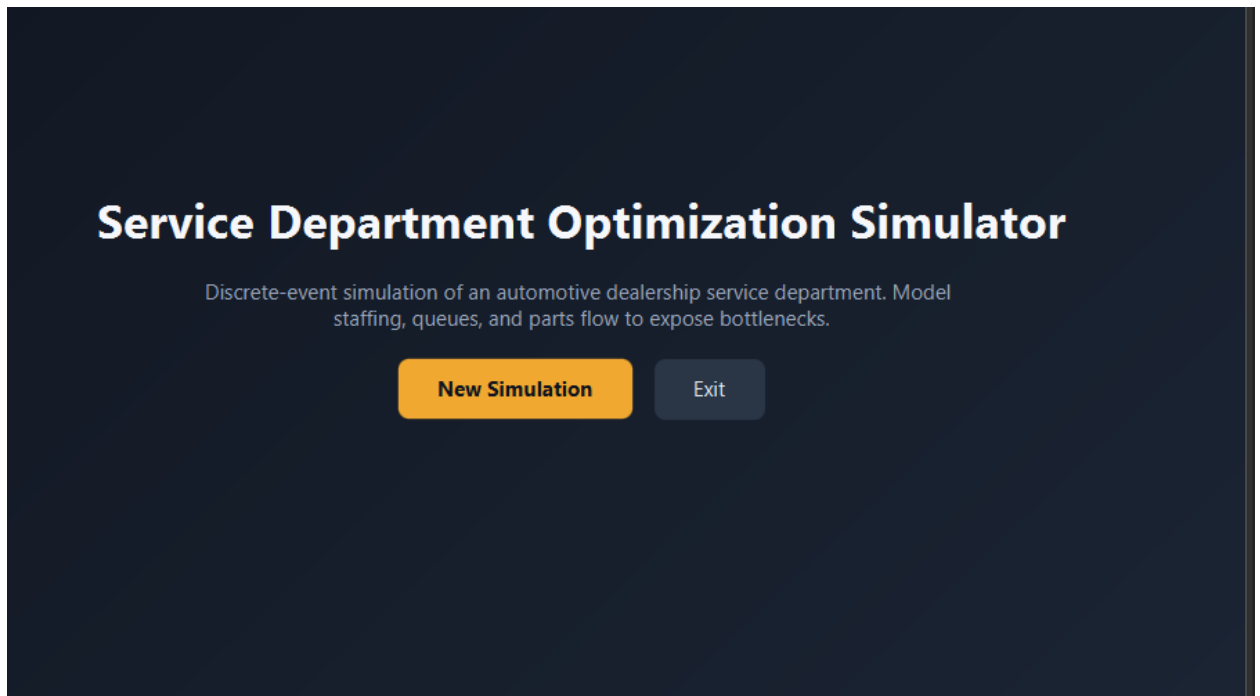
main
├── java
│   └── simulator
│       ├── config
│       │   ├── ArrivalProfile
│       │   ├── ServiceTimeModel
│       │   ├── SimulationConfig
│       │   └── SimulationConfigLoader
│       ├── data
│       │   ├── CsvWriter
│       │   ├── DataRecorder
│       │   ├── EventRecord
│       │   ├── JsonWriter
│       │   ├── RunResultWriter
│       │   └── TimeSeriesSample
│       ├── engine
│       ├── inventory
│       │   ├── PartsDepartment
│       │   ├── PartsFulfillmentResult
│       │   ├── PartsFulfillmentStatus
│       │   ├── PartsInventoryEquations
│       │   └── PendingPartOrder
│       ├── metrics
│       │   ├── BayUtilizationMetrics
│       │   ├── CustomerWaitMetrics
│       │   ├── MetricsCollector
│       │   ├── MetricsEquations
│       │   ├── MetricsReport
│       │   ├── PartsDelayMetrics
│       │   ├── QueueBenchmarkMetrics
│       │   ├── SimulationValidationMetrics
│       │   ├── TechnicianUtilizationMetrics
│       │   ├── ThroughputMetrics
│       │   └── ValidationReport
│       └── model
│           ├── Customer
│           ├── JobQueue
│           ├── Part
│           └── ...

```



UI screenshots

Landing Screen



Configuration Screen

Configure Simulation

Horizon (hours)

Arrival rate (λ , /h)

Technicians

Service rate (μ , /h)

Advisors

Random seed

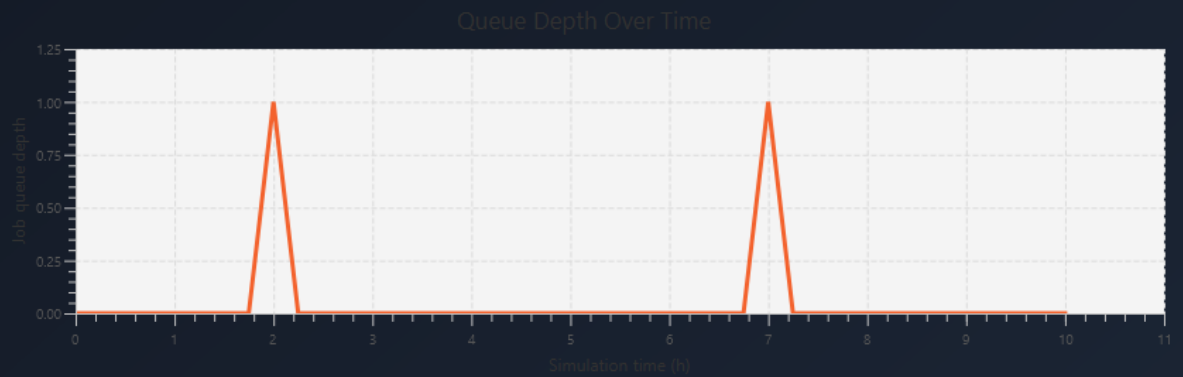
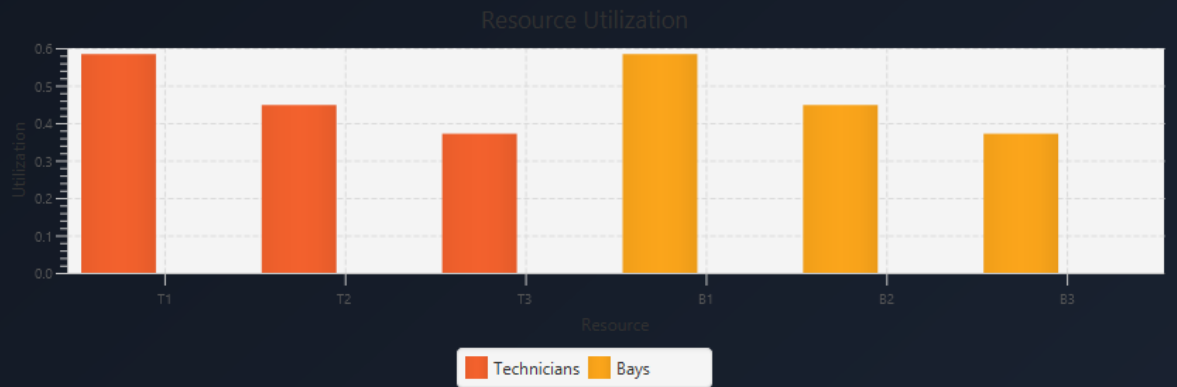
Run Simulation

Back

Run 1 Results

Simulation Results

Jobs completed	41
Avg customer wait (h)	0.393
Avg service time (h)	0.342
Shop technician utilization	0.468
Shop bay utilization	0.468
Analytical rho	0.667
Analytical queue wait W_q (h)	0.222
Run time (ms)	187



Export CSV/JSON

New Run

Home

2. Execution Documentation

Run ID	Purpose	Parameters Changed	Duration (ms)	Status
001	Baseline	Defaults	69	Complete
002	High Arrivals	Arrival rates = 6	24	Complete
003	Low Techs	Technicians = 2	20	Complete
004	Low Advisors	Advisors = 2	23	Complete

005	Fast services	Service Rate = 1	20	Complete
006	Low Arrivals	Arrival rates = 2	7	Complete
007	Understaffing	Techs= 1 Advisors= 3	16	Complete
008	Many Big Jobs	$\Lambda/hr = 6$ $M/hr = 6$	23	Complete
009	Overstaffed	T = 6 A = 6	15	Complete
010	Too many arrivals	Arrival rate = 10	34	Complete

3. Data Collection Overview

a. Output Structure

- i. Master_index.csv- Indexes the runs of a single session
- ii. Run_xxx_timeseries.csv- Describes system state every quarter hour
- iii. Run_xxx_events.csv- Describes system state every run
- iv. Run_xxx_Summary.json- Aggregate of statistics
- v. Run_xxx_config.json- Exact parameters for each run

b. ii. Run_xxx_timeseries.csv

Column	Unit	Meaning
real_epoch_millis	Milliseconds	Wall-clock time when snapshot was taken
sim time hours	Hours	Current simulation clock
intake queue	Count	Customer wait times for an advisor
job queue depth	Count	Ticket wait times for a technician
busy technicians	Count	Technicians actively working
busy bays	Count	Bays being utilized
parts on hand	Count	Parts inventory
jobs completed	Count	Cumulative jobs done

```

real_epoch_millis,sim_time_hours,intake_queue,job_queue_depth,busy_technicians,busy_bays,parts_on_hand,
1783630562539,0.0,0,0,0,0,20,0
1783630562539,0.25,0,0,1,1,20,0
1783630562540,0.5,0,0,2,2,15,1
1783630562541,0.75,0,0,3,3,15,1
1783630562542,1.0,0,0,4,4,15,3
1783630562542,1.25,0,0,4,4,12,4
1783630562543,1.5,0,0,3,3,10,7
1783630562543,1.75,0,0,2,2,7,9
1783630562544,2.0,0,0,2,2,4,11
1783630562544,2.25,0,0,2,2,4,12
1783630562545,2.5,0,0,1,1,4,13
1783630562545,2.75,0,0,1,1,4,15
1783630562546,3.0,0,0,2,2,4,16
1783630562546,3.25,0,0,0,0,4,18
1783630562546,3.5,0,0,0,0,4,19
1783630562547,3.75,0,0,2,2,1,19
1783630562547,4.0,0,0,1,1,1,21
1783630562547,4.25,0,0,0,0,1,22
1783630562547,4.5,0,0,0,0,11,22
1783630562547,4.75,0,0,0,0,11,22
1783630562547,5.0,0,0,0,0,11,22
1783630562548,5.25,0,0,1,1,11,22
1783630562548,5.5,0,0,1,1,11,22
1783630562548,5.75,0,0,1,1,11,23
1783630562549,6.0,0,0,1,1,21,24
1783630562549,6.25,0,0,1,1,21,24
1783630562549,6.5,0,0,0,0,21,25
1783630562549,6.75,0,0,0,0,21,25
1783630562549,7.0,0,0,1,1,19,25
1783630562550,7.25,0,0,3,3,18,25
1783630562550,7.5,0,0,2,2,18,27
1783630562550,7.75,0,0,0,0,18,29

```

c. Run_xxx_events.csv

Column	Meaning
real_epoch_millis	Timestamp
sim_time_hours	Event occurrence in simulation
event_type	Event Category
ticket_id	Ticket ID if applicable (-1 in N/A)
detail	Additional content

```

real_epoch_millis,sim_time_hours,event_type,ticket_id,detail
1783630646931,0.0025791407835509482,ARRIVAL,-1,customer arrival
1783630646931,0.06715470956053025,ARRIVAL,-1,customer arrival
1783630646931,0.09866129677984706,ARRIVAL,-1,customer arrival
1783630646932,0.10257914078355096,INTAKE_COMPLETE,1,ready for servi
1783630646932,0.16022055747033742,ARRIVAL,-1,customer arrival
1783630646932,0.16715470956053025,INTAKE_COMPLETE,2,ready for servi
1783630646932,0.19866129677984706,INTAKE_COMPLETE,3,ready for servi
1783630646932,0.26022055747033745,INTAKE_COMPLETE,4,ready for servi
1783630646932,0.3267224329747649,SERVICE_COMPLETE,2,job complete
1783630646932,0.4064206716823642,SERVICE_COMPLETE,3,job complete
1783630646933,0.49139842603035533,ARRIVAL,-1,customer arrival
1783630646933,0.5019204091547902,SERVICE_COMPLETE,4,job complete
1783630646933,0.5218201341272469,ARRIVAL,-1,customer arrival
1783630646933,0.5418581970585294,SERVICE_COMPLETE,1,job complete
1783630646933,0.5913984260303553,INTAKE_COMPLETE,5,ready for servic
1783630646933,0.6058677777370556,ARRIVAL,-1,customer arrival

```

Data

d. Run_xxx_summary.json

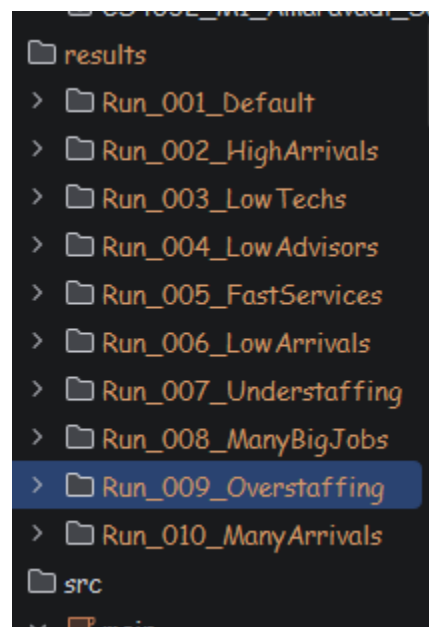
Field	Unit	Meaning
jobsCompletedPerDay	Jobs	Total jobs completed (scaled to day)
averageCustomerWaitHours	Hours	Full customer wait W_{total}
averageAdvisorWaitHours	Hours	Time waiting for advisor
averageQueueDelayHours	Hours	Time in technician queue
averagePartsDelayHours	Hours	Delay from parts unavailability
averageServiceTimeHours	Hours	Active repair time
analyticalQueueWaitHours	Hours	Erlang C expected wait
technicianUtilization	Hashmap	Per-technician utilization
shopTechnicianUtilization	Ratio	Shop-wide tech utilization
shopBayUtilization	Ratio	Shop-wide bay utilization
analyticalSystemUtilization	Ratio	M/M/c theoretical ρ
validation.overallValid	Boolean	PASS/PENDING vs benchmark

```
"runId": 1,  
"purpose": "UI run",  
"wallClockMillis": 34,  
"jobsCompletedPerDay": 100,  
"averageCustomerWaitHours": 1.3127575837430308,  
"averageAdvisorWaitHours": 0.001635161898354004,  
"averageQueueDelayHours": 0.9204938962163635,  
"averagePartsDelayHours": 0.04920680824886505,  
"averageServiceTimeHours": 0.3414217173794482,  
"averageTotalJobDelayHours": 0.9697007044652284,  
"shopTechnicianUtilization": 1.1380723912648278,  
"shopBayUtilization": 1.1380723912648278,  
"analyticalSystemUtilization": 1.6666666666666667,  
"analyticalQueueWaitHours": Infinity,  
"technicianUtilization": {  
  "2": 1.1508089128379027,  
  "3": 1.132051894182553,  
  "1": 1.1313563667740283
```

e. Run_xxx_config.json

```
"randomSeed": 42,  
"simulationHorizonHours": 10.0,  
"arrivalRate": 10.0,  
"arrivalProfile": "DEALERSHIP_DAY",  
"technicianCount": 3,  
"serviceRatePerTechnician": 2.0,  
"advisorCount": 4,  
"customerCount": 0,  
"experienceAlpha": 1.0,  
"maxExperienceLevel": 10,  
"gammaShapeParameter": 4.0,  
"serviceTimeModel": "PDF",  
"partsReorderPoint": 5,  
"partsReorderQuantity": 10,  
"partsLeadTimeHours": 2.0,  
"initialPartsQuantityOnHand": 20,  
"validationRelativeTolerance": 0.15,  
"replicationCount": 1
```

f. Folder Structure



4. Preliminary Results

a. Cross Run Comparison Table

Run	Scenario	Key Params	Jobs done	Avg Cust Wait (hours)	Validation
001	Baseline	Default	41	0.39	Pending
002	Low Arrivals	$\lambda=2$	18	0.38	Pending
003	High Arrivals	$\lambda=6$	60	0.47	Pending
004	Fewer techs	$C = 2$	38	0.90	Pending
005	More Techs	$C = 5$	43	0.40	Pending
006	Slow Day	$\mu=1.0$	41	1.18	Pending
007	Busy Day	$\mu=3.0$	41	0.27	Pending
008	Few Advisors	1 Advisor	41	0.40	Pending
009	Many Advisors	8 Advisors	41	0.39	Pending
010	Scarce Parts	$R = 1$ Qty = 2	41	0.74	Pending
011	Abundant Parts	Qty = 100	41	0.34	Pending
012	Combined Stress	$\lambda=6, c=2, \mu=1$	68	8.98	Pending

This table is incomplete for this milestone, however, will be filled out more significantly for Milestone 4. However initial observations can be made just by an initial parse. Based on initial observations, service times have a large impact on customer wait times while queue and parts delays can make a bad situation worse. Low arrival rates has little impact on waiting times however high arrival rates have an impact on waiting times. Both technician capacity and advisor capacity have a high impact on waiting times.